

Bhargava Factorials for the 2–3 Semigroup

and a New Integer-Valued-Interpolation Frontier

A nonduplicative progress report on the
Alaoglu–Erdős integer-exponent conjecture

$$2^x, 3^x \in \mathbb{Z} \implies x \in \mathbb{Z}$$

Bottom line. No complete proof is claimed. The attached unified report already develops an exceptionally large collection of transcendence, determinant, interpolation, p -adic, dynamical, and formalization arguments. This report deliberately enters the one substantial interface that the supplied corpus names but does not develop: Bhargava p -orderings and generalized factorials.

For the rank-two semigroup

$$\mathcal{B} = \{2^a 3^b : a, b \in \mathbb{N}_0\},$$

we determine the complete local factorial calculus. At every prime $p \geq 5$, if

$$h_p(e) = |\langle 2, 3 \rangle \subset (\mathbb{Z}/p^e \mathbb{Z})^\times|,$$

then

$$v_p(k!_{\mathcal{B}}) = \sum_{e \geq 1} \left\lfloor \frac{k}{h_p(e)} \right\rfloor.$$

At $p = 3$, the semigroup is dense in $\mathbb{Z}_3$, so $v_3(k!_{\mathcal{B}}) = v_3(k!)$. At $p = 2$, the closure splits self-similarly into an odd orbit and a doubled copy of itself; this gives an exact merge recursion and the asymptotic slope

$$\frac{v_2(k!_{\mathcal{B}})}{k} \longrightarrow \frac{\sqrt{11} - 1}{2}.$$

A collision lemma proves that every prime power contributing to $k!_{\mathcal{B}}$ is below $6^{\lfloor \sqrt{k} \rfloor}$, yielding a finite exact algorithm. We also construct the first regular-basis element that cannot arise from one global Newton ordering and determine the optimal normalized Vandermonde defects through eight nodes:

$$1, 1, 1, 1, 1, 7, 7, 396.$$

These local and finite results feed an optimal normalized interpolation determinant. Under a hypothetical nonintegral solution x , the map $q \mapsto q^x$ is integer-valued on $\mathcal{B}$; therefore every regular Bhargava basis produces nonzero integer determinants. This turns the conjecture into a concrete subunit problem in the exact lattice $\text{Int}(\mathcal{B}, \mathbb{Z})$, and also into a finite-support adelic problem after localizing at 2 and 3. The new machinery removes all ambiguity about automatic denominator content. What remains is genuinely Archimedean–nonarchimedean coupling: one must control cancellation or the denominator carried by the structural primes.

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Abstract

The Alaoglu–Erdős conjecture is the assertion that a real number x for which 2^x and 3^x are integers must itself be an integer. If a nonintegral solution existed and $M = 2^x$, $N = 3^x$, then the power map would define an order-preserving monoid isomorphism

$$2^a 3^b \longmapsto M^a N^b$$

between two rank-two integer semigroups. The supplied unified report studies this situation through many complementary frameworks, but leaves Bhargava’s theory of p -orderings as an undeveloped research interface.

This report computes the Bhargava factorial of the input semigroup $\mathcal{B} = \langle 2, 3 \rangle_+$. For odd unramified primes the answer is an exact orbit-size formula; the prime 3 is ordinary because $\mathcal{B}$ is 3-adically dense; and the prime 2 is governed by a self-referential two-branch renewal law whose limiting slope is $(\sqrt{11} - 1)/2$. A finite-support theorem bounds every contributing prime power and makes the factorial exactly computable. We deduce fixed-prime excess formulas, denominator-growth bounds, and a rank-two Artin–Bhargava asymptotic problem.

The factorial calculus is then inserted into integer-valued interpolation. A regular basis of $\text{Int}(\mathcal{B}, \mathbb{Z})$ gives an optimally normalized generalized Vandermonde determinant which, under a counterexample, is a nonzero integer. The first non-Newton basis element is given explicitly, and exhaustive exact certificates yield the defect minima 1, 1, 1, 1, 1, 7, 7, 396 through eight nodes. Equivalently, no degree- d integer-valued polynomial can approximate t^x within one unit at $d + 2$ semigroup nodes. We formulate four finite proof criteria, an S -integer compact-group version, and a Lean-oriented formalization plan. The conjecture remains open: generalized factorials settle the exact local divisibility, but not the output-sensitive cancellation needed to defeat the structural-prime denominator tax.

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1 Scope, trust boundary, and relation to the supplied report

1.1 The mathematical starting point

The conjecture, introduced in the study of highly composite and related numbers [1], is

Conjecture 1.1 (Alaoglu–Erdős). *For $x \in \mathbb{R}$,*

$$2^x \in \mathbb{Z}, \quad 3^x \in \mathbb{Z} \implies x \in \mathbb{Z}.$$

A negative exponent cannot occur, because then $0 < 2^x < 1$. A rational solution is an integer by unique factorization. An algebraic irrational solution is excluded by the Gelfond–Schneider theorem [2]. Thus any counterexample is positive and transcendental. With

$$M = 2^x, \quad N = 3^x,$$

one obtains

$$(2^a 3^b)^x = M^a N^b \in \mathbb{Z} \quad (a, b \in \mathbb{N}_0). \tag{1}$$

The problem is therefore not a two-value accident: a counterexample would make a real power integer-valued on an entire rank-two multiplicative semigroup.

1.2 What is intentionally not repeated

The supplied report is more than twenty-three thousand lines long. It already treats, among many other topics, the four-exponentials barrier, exact conditional semigroup structure, valuation lattices, finite differences, generalized Vandermonde determinants, tropical divisibility ceilings, Sturmian codings, rational-point counting, Loewner matrices, geometric Newton interpolation, q -special functions, Kummer defects, and formalization boundaries. Repeating those arguments would add bulk rather than information.

The present report follows one explicit opening in that corpus. Its arithmetic-interpolation chapter states that broader Bhargava-ordering proposals remain a research interface, but it does not define a p -ordering, compute a generalized factorial, or build the corresponding regular basis. Everything below is organized around that missing local calculus.

1.3 Evidence labels

The labels used here have literal meanings.

[proved here] A complete paper proof is included.

[exact computation] The assertion is the output of exact integer arithmetic; source is included and the mathematical cutoff is proved.

[external theorem] The result is quoted from the literature and is not reproved in full.

[conditional criterion] The statement is a rigorously proved sufficient condition, but its hypothesis has not been established.

[open target] The statement is proposed as a research target or conjectural asymptotic.

No numerical pattern is promoted to a theorem. In particular, the public announcement of a rival Alaoglu–Erdős breakthrough is not available in a citable proof or preprint at the time of writing and is not used.

2 Bhargava factorials and the exact interpolation lattice

2.1 p -orderings

Let $S \subset \mathbb{Z}$ be infinite and let p be a prime. A p -ordering of S is a sequence $a_0, a_1, \dots$ chosen recursively so that, after $a_0, \dots, a_{k-1}$ have been chosen, the value

$$v_p \left(\prod_{i=0}^{k-1} (a - a_i) \right)$$

is minimized over $a \in S$. Bhargava proved that the minimum is independent of the choices made in the ordering [3, 4]. Denote it by

$$\nu_{S,p}(k).$$

The k -th generalized factorial is

$$k!_S = \prod_p p^{\nu_{S,p}(k)}. \quad (2)$$

For $S = \mathbb{Z}$, the ordinary ordering $0, 1, 2, \dots$ gives

$$\nu_{\mathbb{Z},p}(k) = v_p(k!), \quad k!_{\mathbb{Z}} = k!.$$

If $S \subset T$, minimization over the smaller set can only increase the valuation, so $k!_T \mid k!_S$.

The factorial is not merely a divisibility statistic. Define

$$\text{Int}(S, \mathbb{Z}) = \{P \in \mathbb{Q}[T] : P(S) \subset \mathbb{Z}\}.$$

Bhargava’s factorial and characteristic-ideal theory [3, 5, 6, 7] implies that, because $\mathbb{Z}$ is principal, there is a regular basis

$$F_0, F_1, F_2, \dots \in \text{Int}(S, \mathbb{Z}), \quad \deg F_k = k, \quad \text{lc}(F_k) = \frac{1}{k!_S},$$

and

$$\text{Int}_{\leq d}(S, \mathbb{Z}) = \bigoplus_{k=0}^d \mathbb{Z}F_k.$$

Thus $k!_S$ is the optimal leading-coefficient denominator for degree k .

2.2 Closure invariance

The local computation becomes transparent after passing to p -adic closures.

Lemma 2.1 (Closure invariance). *Let $S \subset \mathbb{Z}$ be infinite and let $\overline{S}^p$ be its closure in $\mathbb{Z}_p$. Then*

$$\nu_{S,p}(k) = \nu_{\overline{S}^p,p}(k)$$

for every $k \geq 0$, where the right side is defined by the same greedy rule in the compact set $\overline{S}^p$.

Proof. Fix finitely many previously selected points. The valuation of their difference product is an integer-valued locally constant function away from those points. The minimum over the compact closure is finite because the closure is infinite. A minimizer can be approximated by an element of S closely enough that every difference has the same valuation. Hence the minimum over S is no larger than the minimum over the closure; the reverse inequality is immediate from $S \subset \overline{S}^p$. Induction on the stage gives the claim. $\square$

2.3 A procyclic orbit lemma

Lemma 2.2 (Orbit factorial). *Let $H \subset \mathbb{Z}_p^\times$ be a procyclic closed subgroup, let g be a topological generator, and let*

$$h(e) = |\operatorname{im}(H \rightarrow (\mathbb{Z}/p^e\mathbb{Z})^\times)|.$$

Then $1, g, g^2, \dots$ is a p -ordering of H , and

$$\nu_{H,p}(k) = \sum_{e \geq 1} \left\lfloor \frac{k}{h(e)} \right\rfloor. \quad (3)$$

Proof. For a candidate $y \in H$,

$$v_p \left(\prod_{i < k} (y - g^i) \right) = \sum_{e \geq 1} \# \{0 \leq i < k : y \equiv g^i \pmod{p^e}\}.$$

At level e , the first k powers of g are distributed among the $h(e)$ residue classes with occupancies differing by at most one. Every candidate therefore meets at least $\lfloor k/h(e) \rfloor$ previous points in its class. The candidate g^k attains that minimum simultaneously at every level, because the preceding indices congruent to k modulo $h(e)$ are $k - h(e), k - 2h(e), \dots$. Summing the level occupancies proves the formula. $\square$

Remark 2.3. The formula is local but global in depth. A single number $h(e)$ records the entire two-generator orbit modulo p^e ; no enumeration of semigroup elements is required.

3 The complete local factorial of $\mathcal{B} = \langle 2, 3 \rangle_+$

Set

$$\mathcal{B} = \{2^a 3^b : a, b \in \mathbb{N}_0\}.$$

For a prime $p \geq 5$, define

$$H_{p,e} = \langle 2, 3 \rangle \subset (\mathbb{Z}/p^e\mathbb{Z})^\times, \quad h_p(e) = |H_{p,e}|. \quad (4)$$

3.1 Every unramified odd prime

Theorem 3.1 (Exact unramified formula). *For every prime $p \geq 5$ and every $k \geq 0$,*

$$\boxed{\nu_{\mathcal{B},p}(k) = \sum_{e \geq 1} \left\lfloor \frac{k}{h_p(e)} \right\rfloor.} \quad (5)$$

Moreover,

$$h_p(e) = \operatorname{lcm}(\operatorname{ord}_{p^e}(2), \operatorname{ord}_{p^e}(3)). \quad (6)$$

Proof. Modulo p^e , nonnegative powers of 2 and 3 already contain their inverses, because the ambient group is finite. Hence the image of $\mathcal{B}$ is precisely $H_{p,e}$. The closure of $\mathcal{B}$ in $\mathbb{Z}_p^\times$ is the closed subgroup generated by 2 and 3. For odd p , $\mathbb{Z}_p^\times$ is procyclic, and every closed subgroup of a procyclic group is procyclic. Lemma 2.1 and Lemma 2.2 give (5). Since $(\mathbb{Z}/p^e\mathbb{Z})^\times$ is cyclic, the subgroup generated by two elements has order the least common multiple of their individual orders, proving (6). $\square$

3.2 Prime-power lifting and the local density

The sequence $h_p(e)$ has a simple eventual form. Let $\lambda_p = h_p(1)$. There is an integer $c_p \geq 1$, the saturation depth of the closed subgroup, such that

$$h_p(e) = \lambda_p p^{\max(0, e - c_p)}. \quad (7)$$

For the usual non-Wieferich behavior, $c_p = 1$. The exact formula then becomes

$$\nu_{\mathcal{B},p}(k) = c_p \left\lfloor \frac{k}{\lambda_p} \right\rfloor + \sum_{j \geq 1} \left\lfloor \frac{k}{\lambda_p p^j} \right\rfloor.$$

In particular the fixed-prime slope exists:

$$\alpha_p := \lim_{k \rightarrow \infty} \frac{\nu_{\mathcal{B},p}(k)}{k} = \sum_{e \geq 1} \frac{1}{h_p(e)}. \quad (8)$$

For comparison,

$$\lim_{k \rightarrow \infty} \frac{v_p(k!)}{k} = \frac{1}{p-1}.$$

Because $h_p(e) \leq \varphi(p^e) = (p-1)p^{e-1}$,

$$\alpha_p \geq \frac{p}{(p-1)^2} = \frac{1}{p-1} + \frac{1}{(p-1)^2}. \quad (9)$$

Thus every fixed unramified prime contributes strictly more factorial mass on $\mathcal{B}$ than it does on all of $\mathbb{Z}$.

3.3 The prime 3: no gain

Theorem 3.2 (Three-adic density). *The closure of $\mathcal{B}$ in $\mathbb{Z}_3$ is all of $\mathbb{Z}_3$. Consequently,*

$$\boxed{\nu_{\mathcal{B},3}(k) = v_3(k!).} \quad (10)$$

Proof. The integer 2 is a primitive root modulo 3^e for every $e \geq 1$. Every nonzero residue modulo 3^e has the form $3^b u$, with $0 \leq b < e$ and u a unit modulo 3^{e-b} . Choose a with $2^a \equiv u \pmod{3^{e-b}}$; then $2^a 3^b$ realizes the residue. The zero residue is realized by 3^e . Hence $\mathcal{B}$ is surjective modulo every 3^e , so its closure is $\mathbb{Z}_3$. Closure invariance reduces the factorial to the ordinary one. $\square$

3.4 The prime 2: a self-similar residue tree

The prime 2 is the genuinely new local feature. Let

$$K = \overline{\mathcal{B}}^2 \subset \mathbb{Z}_2, \quad U = \overline{\langle 3 \rangle} \subset \mathbb{Z}_2^\times.$$

Every element of K is either odd or twice another element of K , and the two pieces are separated modulo 2:

$$K = U \sqcup 2K. \quad (11)$$

The orbit sizes of U are

$$|\langle 3 \rangle \bmod 2| = 1, \quad |\langle 3 \rangle \bmod 4| = 2, \quad |\langle 3 \rangle \bmod 2^e| = 2^{e-2} \quad (e \geq 3).$$

Hence Lemma 2.2 gives

$$u_k := \nu_{U,2}(k) = k + \left\lfloor \frac{k}{2} \right\rfloor + v_2(k!). \quad (12)$$

Theorem 3.3 (Exact two-adic merge recursion). *Let*

$$a_k = \nu_{\mathcal{B},2}(k).$$

Set $a_0 = 0$. After selecting the initial odd branch, maintain counts r and s of previously selected points from $2K$ and U , respectively. Initially $r = 0, s = 1$. At stage $k \geq 1$, compare

$$E_r = r + a_r, \quad O_s = u_s. \quad (13)$$

Set $a_k = \min(E_r, O_s)$ and increment the count of a branch attaining the minimum. Any tie rule gives the same value sequence. Equivalently, as multisets,

$$\{a_k : k \geq 0\} = \{j + a_j : j \geq 0\} \uplus \{u_j : j \geq 0\}, \quad (14)$$

with both sides listed in nondecreasing order.

The first values are

$$0, 0, 1, 1, 3, 4, 4, 5, 7, 9, 9, 10, 10, 12, 13, 14, 15, 18, 19, 19, 20, \dots \quad (15)$$

Proof. Differences between an odd point and an even point are 2-adic units, so cross-branch points contribute zero valuation. Within U , the s -th local cost is u_s . Within $2K$, every difference acquires one factor of 2, and after dividing by 2 the local problem is again K ; with r previous even points the cost is therefore $r + a_r$. The global greedy ordering is exactly the merge of these two local cost streams. Independence of the resulting p -sequence from tie choices is Bhargava’s invariance theorem. $\square$

3.5 The renewal slope

Lemma 3.4 (Self-similar merge slope). *Let b_k be nondecreasing with $b_k/k \rightarrow \beta > 0$. Suppose a nondecreasing unbounded sequence a_k is the ordered multiset merge of*

$$\{k + a_k : k \geq 0\} \quad \text{and} \quad \{b_k : k \geq 0\}.$$

Then a_k/k converges to the unique positive c satisfying

$$\frac{1}{c} = \frac{1}{1+c} + \frac{1}{\beta}. \quad (16)$$

Proof. Let $F(t) = \#\{k : a_k \leq t\}$. The merge identity gives

$$F(t) = \#\{k : k + a_k \leq t\} + \#\{k : b_k \leq t\}.$$

The second term is $t/\beta + o(t)$. Put

$$c_- = \liminf_{k \rightarrow \infty} \frac{a_k}{k}, \quad c_+ = \limsup_{k \rightarrow \infty} \frac{a_k}{k}.$$

The merge itself gives finite positive linear bounds, so $0 < c_- \leq c_+ < \infty$. Monotone inversion yields

$$\liminf_{t \rightarrow \infty} \frac{F(t)}{t} = \frac{1}{c_+}, \quad \limsup_{t \rightarrow \infty} \frac{F(t)}{t} = \frac{1}{c_-}.$$

The first count on the right of (3.5) has lower density at least $1/(1 + c_+)$ and upper density at most $1/(1 + c_-)$. Consequently,

$$\frac{1}{c_+} \geq \frac{1}{1 + c_+} + \frac{1}{\beta}, \quad \frac{1}{c_-} \leq \frac{1}{1 + c_-} + \frac{1}{\beta}.$$

The function $c \mapsto 1/[c(1 + c)]$ is strictly decreasing. Both inequalities force $c_+ \leq c \leq c_-$, where c is the unique solution of (16). Hence $c_- = c_+ = c$. $\square$

Since

$$u_k = 2k + \lfloor k/2 \rfloor - s_2(k) = \frac{5}{2}k + O(\log k),$$

where $s_2(k)$ is the binary digit sum, the lemma applies with $\beta = 5/2$.

Corollary 3.5 (Exact two-adic linear constant).

$$\boxed{\lim_{k \rightarrow \infty} \frac{\nu_{\mathcal{B},2}(k)}{k} = \frac{\sqrt{11} - 1}{2} = 1.1583123951777\dots} \quad (17)$$

Proof. Equation (16) becomes

$$\frac{1}{c} = \frac{1}{1 + c} + \frac{2}{5}, \quad 2c(1 + c) = 5.$$

The positive root is $(\sqrt{11} - 1)/2$. $\square$

Status. The formulas in this section are [\[proved here\]](#). They use only elementary p -adic topology, cyclic unit groups at odd primes, the order of 3 modulo powers of 2, and Bhargava’s choice-independence theorem. They are new relative to the supplied corpus. No claim of priority over unpublished or unindexed work is made.

4 Global factorization and a finite exact algorithm

4.1 A collision cutoff for every contributing prime power

Formula (5) is formally infinite, but for each k only finitely many prime powers can occur. The following bound is deliberately elementary and proof-producing.

Theorem 4.1 (Square-box cutoff). *Let $k \geq 2$, put $R = \lfloor \sqrt{k} \rfloor$, and let $p \geq 5$ be prime. If $h_p(e) \leq k$, then*

$$\boxed{p^e < 6^R.} \quad (18)$$

Proof. The $(R+1)^2$ semigroup elements

$$2^a 3^b, \quad 0 \leq a, b \leq R,$$

map into a group of cardinality $h_p(e) \leq k < (R+1)^2$. Two distinct exponent pairs therefore collide modulo p^e . Unique factorization makes the corresponding integers distinct, so p^e divides a nonzero difference. Both integers lie between 1 and 6^R ; hence the absolute difference is less than 6^R . $\square$

Remark 4.2. A simplex of exponent pairs gives slightly sharper constants, but the square cutoff has three advantages: it is one line, it is uniform in the prime and lifting depth, and it makes exact verification straightforward. The bound is not intended as an asymptotic estimate for the largest prime factor; it is a completeness certificate for the computation.

4.2 The complete factorization theorem

Theorem 4.3 (Bhargava factorial of the 2–3 semigroup). *For every $k \geq 0$,*

$$\boxed{k!_{\mathcal{B}} = 2^{a_k} 3^{v_3(k!)} \prod_{p \geq 5} p^{\sum_{e \geq 1} \lfloor k/h_p(e) \rfloor}}. \quad (19)$$

where a_k is the exact merge sequence of Theorem 3.3. For $k \geq 2$, every prime power that contributes to the product satisfies $p^e < 6^{\lfloor \sqrt{k} \rfloor}$.

Proof. Combine Theorems 3.1, 3.2, and 3.3. The cutoff is Theorem 4.1. $\square$

This gives a finite exact procedure:

1. compute a_k by the two-branch merge;
2. add $v_3(k!)$;
3. enumerate primes $5 \leq p < 6^R$;
4. for powers $p^e < 6^R$, compute

$$h_p(e) = \text{lcm}(\text{ord}_{p^e}(2), \text{ord}_{p^e}(3));$$

5. add $\lfloor k/h_p(e) \rfloor$ to the exponent of p .

No floating-point arithmetic enters.

4.3 First exact values

Table 1 separates the structural exponents from the unramified factor. The last column is the exact integer gain over the ordinary factorial.

Table 1: Exact values of $k!_{\mathcal{B}}$ through $k = 20$.

k	$v_2(k!_{\mathcal{B}})$	$v_3(k!_{\mathcal{B}})$	$\prod_{p \geq 5} p^{v_p(k!_{\mathcal{B}})}$	$k!_{\mathcal{B}}/k!$
0	0	0	1	1
1	0	0	1	1
2	1	0	1	1
3	1	1	1	1
4	3	1	5	5
5	4	1	5	2
6	4	2	5 7	7
7	5	2	5 7	2
8	7	2	5 ² 7	5
9	9	4	5 ² 7	20
10	9	4	5 ² 7 11	22
11	10	4	5 ² 7 11 23	92
12	10	5	5 ³ 7 ² 11 13 23	10465
13	12	5	5 ³ 7 ² 11 13 23	3220
14	13	5	5 ³ 7 ² 11 13 23	460
15	14	6	5 ³ 7 ² 11 13 23	184
16	15	6	5 ⁴ 7 ² 11 13 17 23	1955
17	18	6	5 ⁴ 7 ² 11 13 17 23	920
18	19	8	5 ⁴ 7 ³ 11 13 17 19 23	122360
19	19	8	5 ⁴ 7 ³ 11 13 17 19 23	6440
20	20	8	5 ⁶ 7 ³ 11 ² 13 17 19 23	177100

Several features appear immediately.

- The first proper-orbit prime is 23: both 2 and 3 have order 11 modulo 23, so $h_{23}(1) = 11$, and $23 \mid 11!_{\mathcal{B}}$.
- The ratio $k!_{\mathcal{B}}/k!$ is an integer but need not be monotone. A generalized factorial is indexed by degree, not generated by multiplying successive ratios.
- Exceptional orbit primes can be much larger than k . The prime 431, for example, has $h_{431}(1) = 43$ and first enters at degree 43.

Selected larger values are shown in Table 2.

4.4 Independent consistency checks

The implementation in Appendix A performs the following checks.

1. It verifies every reported multiplicative order by exact modular exponentiation.

k	v_2	v_3	largest prime	support size	$k!_{\mathcal{B}}/k!$	$\log(k!_{\mathcal{B}}/k!)$
20	20	8	23	9	177100	12.084470
25	25	10	47	10	3934840	15.185381
30	31	14	47	12	412855520	19.838608
35	38	15	71	13	564887360	20.152137
40	43	18	73	16	1627306323412000	35.025702
45	48	21	431	18	1001511424282768000	41.448042
50	53	22	431	19	146685344363879921600	46.434821

Table 2: Selected exact factorial diagnostics. “Support size” counts all prime divisors.

2. It recomputes the 2-adic stream from the residue-tree merge, rather than from a fitted formula.
3. For small k and small primes, it compares the orbit formula with direct greedy minimization over a large finite rectangle of exponent pairs.
4. It verifies $k! \mid k!_{\mathcal{B}}$, a consequence of $\mathcal{B} \subset \mathbb{Z}$.
5. It refuses to omit a prime power unless Theorem 4.1 places it beyond the complete search range.

Status. The factorization theorem and the cutoff are [\[proved here\]](#). Tables 1 and 2 are [\[exact computation\]](#) with the source printed verbatim in Appendix A.

5 Local excess, global growth, and a rank-two Artin problem

5.1 Fixed-prime excess over the ordinary factorial

For a fixed prime, floors introduce only logarithmic error. Define

$$\delta_2 = \frac{\sqrt{11} - 1}{2} - 1 = \frac{\sqrt{11} - 3}{2}, \quad \delta_3 = 0,$$

and for $p \geq 5$,

$$\delta_p = \sum_{e \geq 1} \frac{1}{h_p(e)} - \frac{1}{p-1}. \quad (20)$$

Proposition 5.1 (Local excess law). *For every fixed prime p ,*

$$v_p(k!_{\mathcal{B}}) - v_p(k!) = \delta_p k + o_p(k). \quad (21)$$

For $p = 3$ the error is zero, and for every fixed $p \geq 5$ it is $O_p(\log k)$. Moreover,

$$\delta_2 > 0, \quad \delta_3 = 0, \quad \delta_p \geq \frac{1}{(p-1)^2} > 0 \quad (p \geq 5).$$

Proof. At $p = 2$, Corollary 3.5 gives the stated $o(k)$ remainder after subtracting $v_2(k!) = k + O(\log k)$. At $p = 3$, Theorem 3.2 is exact. At $p \geq 5$, sum $\lfloor k/h_p(e) \rfloor = k/h_p(e) + O(1)$ over the $O_p(\log k)$ levels for which $h_p(e) \leq k$, and compare with Legendre’s formula. Inequality (9) gives the last assertion. $\square$

Consequently, for every finite set P of primes containing 2 but not 3,

$$\log \frac{k!_{\mathcal{B}}}{k!} \geq k \sum_{p \in P} \delta_p \log p + o_P(k). \quad (22)$$

Already $p = 2, 5, 7, 11, 13, 17, 19, 23$ give a certified linear coefficient exceeding 0.48. This is a genuine denominator gain, not a numerical artifact.

5.2 Why the global asymptotic is harder

One might expect a Stirling law

$$\log k!_{\mathcal{B}} = k \log k + C_{\mathcal{B}} k + o(k). \quad (23)$$

The fixed-prime analysis is compatible with such a formula, but does not prove it. The issue is the moving set of primes $p > k$ for which the subgroup $\langle 2, 3 \rangle \bmod p$ has unexpectedly small order. Each such prime contributes to $k!_{\mathcal{B}}$ even though it does not divide $k!$.

Research problem 5.2 (Rank-two Artin–Bhargava constant). *Prove or disprove the existence of a constant $C_{\mathcal{B}}$ satisfying (23), and express it as a convergent combination of:*

1. *the two structural local constants at 2 and 3;*
2. *the orbit densities $\sum_e 1/h_p(e)$ at unramified primes;*
3. *a centered correction for exceptional small-order primes.*

This problem is adjacent to, but not identical with, Artin’s primitive-root conjecture. Primitive-root questions ask whether a reduction has maximal order. Here every possible orbit size is weighted by its reciprocal, and prime-power lifting enters as well. Average-order theorems for finitely generated subgroups and GRH-conditional Artin-density formulas are natural inputs, but a uniform summation strong enough for (23) still has to be written down.

5.3 Unconditional growth bounds

The ordinary factorial gives the lower bound

$$\log k! \leq \log k!_{\mathcal{B}}. \quad (24)$$

A first upper bound comes from subset monotonicity, not from an arbitrary star product. Let $D_2 = \{2^n : n \in \mathbb{N}_0\}$. Since $D_2 \subset \mathcal{B}$, one has $k!_{\mathcal{B}} \mid k!_{D_2}$. The natural geometric ordering is a simultaneous local ordering of D_2 , so

$$k!_{D_2} = \prod_{j=0}^{k-1} (2^k - 2^j).$$

Consequently

$$\log k!_{\mathcal{B}} \leq k^2 \log 2 + O(k). \quad (25)$$

The rank-two geometry improves the exponent, but a cumulative-energy argument is required. For

$$N(L) = \#\{(a, b) \in \mathbb{N}_0^2 : a \log 2 + b \log 3 \leq L\},$$

elementary lattice counting gives

$$N(L) = \frac{L^2}{2 \log 2 \log 3} + O(L).$$

Write

$$D_m(\mathcal{B}) = \prod_{j=0}^{m-1} j!_{\mathcal{B}}.$$

Energy divisibility, proved later in Proposition 7.2, implies that $D_m(\mathcal{B})$ divides the Vandermonde product of any m semigroup nodes. Also the local sequences $\nu_{\mathcal{B},p}(k)$ are nondecreasing: the first k factors in the defining product for stage $k+1$ already have valuation at least $\nu_{\mathcal{B},p}(k)$. Hence $k!_{\mathcal{B}} \mid j!_{\mathcal{B}}$ whenever $j \geq k$.

Proposition 5.3 (Rank-two cumulative-energy upper bound). *Put*

$$C_* = \frac{3}{4} \left(\frac{5}{3}\right)^{5/2} \sqrt{2 \log 2 \log 3} = 3.3191974798 \dots$$

Then

$$\log k!_{\mathcal{B}} \leq C_* k^{3/2} + O(k). \quad (26)$$

Proof. Fix $c > 1$, let $m = \lceil ck \rceil$, and choose m semigroup elements below e^L , where

$$L = \sqrt{2 \log 2 \log 3} \sqrt{m} + O(1).$$

Their Vandermonde product is at most $e^{L \binom{m}{2}}$, so energy divisibility gives

$$\log D_m(\mathcal{B}) \leq L \binom{m}{2}.$$

Because every factor $j!_{\mathcal{B}}$ with $k \leq j < m$ is a multiple of $k!_{\mathcal{B}}$,

$$(m - k) \log k!_{\mathcal{B}} \leq \log D_m(\mathcal{B}).$$

It follows that

$$\log k!_{\mathcal{B}} \leq \left(\frac{c^{5/2}}{2(c-1)} \sqrt{2 \log 2 \log 3} + o(1) \right) k^{3/2}.$$

The coefficient is minimized at $c = 5/3$, giving C_* . The lattice-counting and rounding errors contribute $O(k)$. $\square$

The use of cumulative energy is essential. In a sparse set, $k!_S$ need not divide the product of the differences from an arbitrary node to k arbitrary companions; for example, this already fails on $\mathcal{B}$ at degree six. What is universal is the full Vandermonde divisor $D_m(S)$. The gap between $k \log k$ and $k^{3/2}$ remains substantial. Closing it is not required for the Alaoglu–Erdős conjecture, but a linear-error Stirling law would make the normalized interpolation lattice quantitatively transparent.

5.4 Recent determinant-completion context

Bhargava’s factorials were created to organize fixed divisors and integer-valued polynomial bases. Recent work studies Stirling laws for polynomial images, simultaneous p -orderings, and determinant-like completions of generalized factorials [9, 8, 10]. In particular, a July 2026 working draft proposes renewal and determinant-line models and explicitly separates the multivariable theory as a further problem. The semigroup $\mathcal{B}$ is a natural rank-two test: its odd-prime pieces are procyclic orbit systems, its 3-adic piece is ordinary, and its 2-adic piece is a solvable nonlinear renewal. This report uses only the elementary local consequences, not any conjectural analytic completion.

Status. Proposition 5.1 and the growth bounds are [\[proved here\]](#). Problem 5.2 is [\[open target\]](#). Its formulation is motivated by the exact factorial calculus, not assumed in any later argument.

6 A counterexample as an isomorphism of two factorial semigroups

Assume for this section that x is a hypothetical nonintegral solution and put

$$M = 2^x, \quad N = 3^x.$$

Define the output semigroup

$$\mathcal{C} = \{M^a N^b : a, b \in \mathbb{N}_0\}.$$

Proposition 6.1 (Power-semigroup isomorphism). *The map*

$$\Phi_x : \mathcal{B} \longrightarrow \mathcal{C}, \quad \Phi_x(2^a 3^b) = M^a N^b = (2^a 3^b)^x$$

is an order-preserving monoid isomorphism. Both $2, 3$ and M, N are multiplicatively independent pairs.

Proof. Multiplicative independence of $2, 3$ is unique factorization. If $M^r = N^s$, then $2^{xr} = 3^{xs}$; because $x > 0$, this implies $2^r = 3^s$, hence $r = s = 0$. Thus exponent pairs give unique coordinates on both semigroups. Multiplicativity is immediate, and $t \mapsto t^x$ is strictly increasing on $\mathbb{R}_{>0}$. $\square$

This reformulation separates two geometries.

Archimedean geometry. Φ_x preserves the total ordering and raises all ratios to the power x .

Finite-place geometry. The residue trees of $\langle 2, 3 \rangle$ and $\langle M, N \rangle$ can be completely different. At a prime not dividing the generators, each is controlled by the size of a generated subgroup; at a prime dividing a generator, a ramified branch tree appears.

The exact input factorial is now known. The output factorial has the same formal local description: for every odd prime $p \nmid MN$,

$$v_p(k!_{\mathcal{C}}) = \sum_{e \geq 1} \left\lfloor \frac{k}{|\langle M, N \rangle \bmod p^e|} \right\rfloor. \quad (27)$$

At primes dividing MN , the closure decomposes into valuation branches, just as $\mathcal{B}$ decomposes at 2. This makes the pair

$$(k!_{\mathcal{B}}, k!_{\mathcal{C}})_{k \geq 0}$$

a new invariant of a hypothetical solution. The order isomorphism controls the real sizes of the elements but does not, by itself, identify these two factorial sequences.

Research problem 6.2 (Factorial-spectrum rigidity). *Find a property of an order-preserving power isomorphism $\Phi_x : \mathcal{B} \rightarrow \mathcal{C}$ which forces a quantitative relation between $k!_{\mathcal{B}}$ and $k!_{\mathcal{C}}$ strong enough to imply that x is an integer.*

A naive equality of factorials is false and should not be expected: Bhargava factorials are arithmetic invariants of the embedded sets, not abstract monoid invariants. Any successful comparison has to use the special analytic form $q \mapsto q^x$.

7 The optimally normalized interpolation determinant

7.1 An integer determinant forced by a counterexample

Fix a regular basis $F_0, F_1, \dots$ of $\text{Int}(\mathcal{B}, \mathbb{Z})$ as in (2.1). For $d \geq 0$ and distinct nodes

$$E = (q_0, \dots, q_{d+1}) \subset \mathcal{B},$$

define

$$\Delta_{d,x}(E) = \det \begin{pmatrix} F_0(q_0) & F_1(q_0) & \cdots & F_d(q_0) & q_0^x \\ F_0(q_1) & F_1(q_1) & \cdots & F_d(q_1) & q_1^x \\ \vdots & \vdots & & \vdots & \vdots \\ F_0(q_{d+1}) & F_1(q_{d+1}) & \cdots & F_d(q_{d+1}) & q_{d+1}^x \end{pmatrix}. \quad (28)$$

Theorem 7.1 (Bhargava-normalized determinant). *Under a hypothetical nonintegral solution x ,*

$$\boxed{\Delta_{d,x}(E) \in \mathbb{Z} \setminus \{0\}}. \quad (29)$$

Moreover,

$$\Delta_{d,x}(E) = \frac{\det(q_i^{\lambda_j})_{0 \leq i,j \leq d+1, (\lambda_0, \dots, \lambda_{d+1}) = (0, 1, \dots, d, x)}}{\prod_{j=0}^d j!_{\mathcal{B}}}. \quad (30)$$

Proof. Every $F_j(q_i)$ is an integer, and every q_i^x is an integer by (1); hence the determinant is integral. It is nonzero because the functions

$$1, t, t^2, \dots, t^d, t^x$$

form a Chebyshev system on $\mathbb{R}_{>0}$ when $x \notin \{0, 1, \dots, d\}$. Equivalently, after the change $t = e^y$, a nontrivial linear combination is an exponential polynomial with distinct real exponents and has at most $d + 1$ real zeros.

The change of columns from $1, T, \dots, T^d$ to the regular basis is upper triangular with diagonal entries $1/j!_{\mathcal{B}}$. Its determinant is $\prod_{j=0}^d (j!_{\mathcal{B}})^{-1}$, proving (30). $\square$

7.2 Divided differences and an exact inequality

Let

$$V(E) = \prod_{0 \leq i < j \leq d+1} (q_j - q_i)$$

with the nodes listed increasingly. The standard divided-difference identity gives

$$\det(1, q_i, \dots, q_i^d, q_i^x) = V(E) [q_0, \dots, q_{d+1}] t^x. \quad (31)$$

For some $\xi \in (q_0, q_{d+1})$,

$$[q_0, \dots, q_{d+1}] t^x = \binom{x}{d+1} \xi^{x-d-1}. \quad (32)$$

Combining this with Theorem 7.1 gives a necessary inequality for every counterexample:

$$\boxed{\frac{|V(E)|}{\prod_{j=0}^d j!_{\mathcal{B}}} \left| \binom{x}{d+1} \right| \xi^{x-d-1} \geq 1.} \quad (33)$$

This is the exact place where the generalized factorial matters. Replacing $\prod j!_{\mathcal{B}}$ by ordinary factorials throws away genuine local divisibility; replacing it by a node-specific product obscures which part is universal and which part is geometric.

7.3 The factorial defect of a node set

Applying the regular basis through degree $m-1$ to m nodes gives:

Proposition 7.2 (Energy divisibility). *For every m -element set $E \subset \mathcal{B}$,*

$$\prod_{j=0}^{m-1} j!_{\mathcal{B}} \mid V(E). \quad (34)$$

Consequently,

$$Q_{\mathcal{B}}(E) := \frac{|V(E)|}{\prod_{j=0}^{m-1} j!_{\mathcal{B}}} \in \mathbb{N} \quad (35)$$

is a well-defined factorial defect.

Proof. The determinant of the integer matrix $(F_j(q_i))_{0 \leq i, j < m}$ is an integer. The triangular leading-coefficient transformation identifies that determinant with $V(E) / \prod_{j=0}^{m-1} j!_{\mathcal{B}}$, up to sign. $\square$

A set with $Q_{\mathcal{B}}(E) = 1$ is simultaneously optimal at every prime. Such sets are finite prefixes of a simultaneous p -ordering. In terms of $Q_{\mathcal{B}}$, (33) becomes

$$Q_{\mathcal{B}}(E) (d+1)!_{\mathcal{B}} \left| \binom{x}{d+1} \right| \xi^{x-d-1} \geq 1. \quad (36)$$

This isolates three independent quantities: the universal factorial, the geometric defect of the chosen nodes, and the Archimedean curvature of t^x .

8 The first global incompatibility: an unavoidable factor 7

The first few semigroup points behave deceptively like an ordinary Newton sequence:

$$\{1, 2\}, \quad \{1, 2, 3\}, \quad \{1, 2, 3, 4\}, \quad \{1, 2, 3, 4, 6\}$$

have factorial defect 1. The next step fails.

Theorem 8.1 (First factorial defect). *For every six-element subset $E \subset \mathcal{B}$,*

$$Q_{\mathcal{B}}(E) \geq 7. \quad (37)$$

Equality is attained by

$$E_6 = \{1, 2, 3, 4, 6, 8\}, \quad Q_{\mathcal{B}}(E_6) = 7. \quad (38)$$

Thus $\mathcal{B}$ has no simultaneous p -ordering whose first six points are globally optimal.

Proof. The exact denominator is

$$\prod_{j=0}^5 j!_{\mathcal{B}} = 1 \cdot 1 \cdot 2 \cdot 6 \cdot 120 \cdot 240 = 345600.$$

Translate a six-point integer set so its least point is 0, writing

$$0 = y_0 < y_1 < \cdots < y_5 = L.$$

Its Vandermonde product is translation-invariant. The four interior points contribute at least the ordinary four-point Vandermonde $1!2!3! = 12$, and $y_i(L - y_i) \geq L - 1$. Hence

$$V(y_0, \dots, y_5) \geq 12L(L - 1)^4. \quad (39)$$

If $Q_{\mathcal{B}}(E) < 7$, then $V(E) < 7 \cdot 345600$, so (39) gives $L \leq 12$. There are only $\binom{12}{5} = 792$ normalized gap patterns to inspect.

Exact enumeration leaves the following patterns below the threshold, up to reflection:

gap pattern	$V/345600$
(0, 1, 2, 3, 4, 5)	1/10
(0, 1, 2, 3, 4, 6)	3/5
(0, 1, 2, 3, 5, 6)	3/2
(0, 1, 2, 4, 5, 6)	2
(0, 1, 2, 3, 4, 7)	21/10
(0, 2, 3, 4, 5, 7)	7/2
(0, 1, 2, 3, 4, 8)	28/5

together with reflected duplicates. Because Proposition 7.2 requires the ratio to be an integer for a subset of $\mathcal{B}$, the only possible defect below 7 is 2, with gap pattern

$$(0, 1, 2, 4, 5, 6).$$

A translate of this pattern would contain two triples of consecutive 3-smooth integers:

$$t, t + 1, t + 2 \quad \text{and} \quad t + 4, t + 5, t + 6.$$

The only triples of consecutive elements of $\mathcal{B}$ are 1, 2, 3 and 2, 3, 4. Indeed, if the first and third are odd, they are powers of 3 differing by 2, forcing 1, 3. If the middle one is odd, it is 3^a , while its neighbors are not divisible by 3 and must be powers of 2; two powers of 2 differing by 2 force 2, 4. No translate contains two such triples four units apart. Therefore a defect below 7 is impossible.

Finally,

$$V(1, 2, 3, 4, 6, 8) = 7 \cdot 345600,$$

which proves sharpness. The finite pattern list is independently reproduced by the exact script in Appendix A. $\square$

Remark 8.2. The factor 7 is not inserted by hand. The set E_6 is locally optimal at 2, 3, 5; its sole excess prime is 7, coming from the endpoint difference $8 - 1$. The theorem is the first place where primewise optimal orderings cannot be synchronized globally.

Corollary 8.3 (No six-node Newton prefix). *There is no sequence $a_0, \dots, a_5 \in \mathcal{B}$ which is a p -ordering prefix for every prime p simultaneously.*

Caution 8.4. The absence of a simultaneous ordering does not prevent a regular basis of $\text{Int}(\mathcal{B}, \mathbb{Z})$. The regular basis is assembled from compatible characteristic ideals, not from one universal Newton sequence. Confusing these two facts would incorrectly discard the Bhargava lattice just when its first nontrivial global obstruction appears.

Status. The normalized determinant, energy divisibility, and Theorem 8.1 are [\[proved here\]](#). The 792-pattern check is finite exact arithmetic; the source and threshold proof are included, so no heuristic search boundary is involved.

9 Subunit locking in the exact Bhargava lattice

9.1 The optimal integer-valued approximation statement

Theorem 9.1 (Bhargava locking). *Assume $x \notin \mathbb{Z}$ and $q^x \in \mathbb{Z}$ for every $q \in \mathcal{B}$. Let $d \geq 0$ and let $q_0, \dots, q_{d+1}$ be distinct elements of $\mathcal{B}$. Then*

$$\inf_{P \in \text{Int}_{\leq d}(\mathcal{B}, \mathbb{Z})} \max_{0 \leq i \leq d+1} |q_i^x - P(q_i)| \geq 1. \quad (40)$$

Proof. For $P \in \text{Int}(\mathcal{B}, \mathbb{Z})$, every residual $q_i^x - P(q_i)$ is an integer. If all residuals had absolute value below 1, they would all vanish. The nonzero generalized polynomial $t^x - P(t)$ would then have $d + 2$ positive zeros, contradicting the Chebyshev property of $1, t, \dots, t^d, t^x$. $\square$

This is an exact closest-vector obstruction. In a regular basis write

$$P = \sum_{j=0}^d c_j F_j, \quad c_j \in \mathbb{Z}.$$

For $E = (q_0, \dots, q_{d+1})$, let

$$A_E = (F_j(q_i)) \in \mathbb{Z}^{(d+2) \times (d+1)}, \quad y_x(E) = (q_i^x) \in \mathbb{Z}^{d+2}.$$

Then Theorem 9.1 is

$$\text{dist}_{\ell^\infty}(y_x(E), A_E \mathbb{Z}^{d+1}) \geq 1. \quad (41)$$

The matrix A_E is the optimal integer-valued-polynomial lattice; no larger universal denominators are available.

9.2 A determinant dual certificate

The left kernel of A_E has rank one. Let $\lambda(E) \in \mathbb{Z}^{d+2}$ be its primitive generator, with alternating signs for increasing nodes. Then

$$\lambda(E)^T y_x(E) \in \mathbb{Z} \setminus \{0\}, \quad |\lambda(E)^T y_x(E)| \geq 1. \quad (42)$$

This is the cofactor form of Theorem 7.1. It turns the search for a contradiction into a finite task: construct E for which the exact integer linear form has a rigorous Archimedean upper bound below one.

9.3 Four concrete proof criteria

Proof criterion 9.2 (Normalized determinant criterion). *Conjecture 1.1 follows if, for every sufficiently large possible counterexample x , one can find d and $E \subset \mathcal{B}$, $|E| = d + 2$, such that*

$$\left| \det(1, q_i, \dots, q_i^d, q_i^x) \right| < \prod_{j=0}^d j!_{\mathcal{B}}. \quad (43)$$

Proof. The normalized determinant would be a nonzero integer of absolute value below one. $\square$

Proof criterion 9.3 (Low-defect curvature criterion). *It is enough to find d, E , and a valid divided-difference bound such that*

$$Q_{\mathcal{B}}(E)(d+1)!_{\mathcal{B}} \left| \binom{x}{d+1} \right|_{\max_{t \in [\min E, \max E]} t^{x-d-1}} < 1. \quad (44)$$

Proof criterion 9.4 (Regular-basis closest vector). *It is enough to exhibit $P \in \text{Int}_{\leq d}(\mathcal{B}, \mathbb{Z})$ and $d + 2$ semigroup nodes on which*

$$|q^x - P(q)| < 1.$$

The coefficients of P may be searched directly in the regular basis; no interpolation node set is required.

Proof criterion 9.5 (Output-input paired lattice). *Let $G_0, G_1, \dots$ be a regular basis for $\text{Int}(\mathcal{C}, \mathbb{Z})$. It is enough to combine the input determinant for t^x and the output determinant for $t^{1/x}$ into a nonzero rational number whose valuations are nonnegative at every finite prime and whose Archimedean absolute value is below one.*

The fourth criterion is deliberately phrased adelically. Separate real estimates for a function and its inverse do not automatically multiply to something small; the finite-place normalizations must be coordinated. This is a precise cancellation problem, not a missing factorial calculation.

9.4 Why the new lattice is useful even before it closes

The exact factorial performs three reductions.

1. It replaces a vague search over rational polynomial coefficients by the canonical lattice $\text{Int}_{\leq d}(\mathcal{B}, \mathbb{Z})$.
2. It identifies the maximal automatic denominator prime by prime, so any further gain must come from cancellation, special node geometry, or the output values.

3. It supplies short finite certificates: a factorization of $j!_{\mathcal{B}}$, an integer evaluation matrix, and one determinant or dual cofactor vector.

The same facts also explain the limit. Generalized factorials are phase-blind. They know how many previous nodes occupy a residue class, but not the signed cancellation among determinant terms. A proof must add information not contained in the local minima alone.

10 Localizing the semigroup: dense Archimedean orbits and a finite-prime tax

10.1 The full 2–3 group over $\mathbb{Z}[1/6]$

The positive group

$$\Gamma_{2,3} = \{2^a 3^b : a, b \in \mathbb{Z}\}$$

is dense in $\mathbb{R}_{>0}$, because $a \log 2 + b \log 3$ is dense in $\mathbb{R}$. It is naturally a subset of the Dedekind domain

$$R_6 = \mathbb{Z}[1/6].$$

The nonzero prime ideals of R_6 correspond to rational primes $p \geq 5$. Bhargava’s construction therefore gives a localized factorial

$$k!_{\Gamma_{2,3}, R_6} = \prod_{p \geq 5} p^{e \geq 1} \sum [k/h_p(e)] \quad \text{up to a unit of } R_6. \quad (45)$$

The proof is exactly Theorem 3.1: negative exponents do not change the finite image, and the structural primes are inverted.

This version is conceptually clean. All compact unramified orbit data are retained, while the two noncompact valuation directions are isolated as units.

10.2 What a counterexample gives after localization

Under a hypothetical solution,

$$\Phi_x(\Gamma_{2,3}) = \Gamma_{M,N} := \{M^a N^b : a, b \in \mathbb{Z}\} \subset \mathbb{Z}[1/(MN)].$$

Let

$$A = \mathbb{Z}[1/(6MN)].$$

Choose a regular basis $F_0^\times, F_1^\times, \dots$ of $\text{Int}(\Gamma_{2,3}, R_6)$. For distinct $q_0, \dots, q_{d+1} \in \Gamma_{2,3}$, define

$$\Delta_{d,x}^\times(E) = \det(F_0^\times(q_i), \dots, F_d^\times(q_i), q_i^x).$$

The same Chebyshev argument gives

$$\Delta_{d,x}^\times(E) \in A \setminus \{0\}. \quad (46)$$

At every prime $p \nmid 6MN$, its p -adic valuation is nonnegative. Thus every possible denominator is supported on the finite set

$$S = \{p : p \mid 6MN\}.$$

For a nonzero rational z integral outside S , define its denominator budget

$$\mathfrak{d}_S(z) = \prod_{p \in S} p^{\max(0, -v_p(z))}.$$

Then

$$|z|_\infty \geq \frac{1}{\mathfrak{d}_S(z)}. \quad (47)$$

Consequently:

Proof criterion 10.1 (Finite-support compact-orbit criterion). *Conjecture 1.1 follows if one can construct localized node sets E for which*

$$|\Delta_{d,x}^\times(E)|_\infty \mathfrak{d}_S(\Delta_{d,x}^\times(E)) < 1. \quad (48)$$

All primes outside S are already handled optimally by the generalized factorial. The criterion asks only for output-sensitive control at finitely many structural primes.

10.3 Why density alone loses

Take a convergent a/b to $\log_2 3$ and put

$$u = \frac{2^a}{3^b}.$$

Then $u \rightarrow 1$, and for fixed d the geometric cluster

$$E_u = \{1, u, u^2, \dots, u^{d+1}\}$$

has

$$|V(E_u)| \asymp_d |\log u|^{\binom{d+2}{2}}. \quad (49)$$

Continued fractions give $|\log u| \ll 1/b$, so the Archimedean determinant decays only polynomially in the exponent height.

The structural denominator grows exponentially. In the monomial determinant, the entry $(u^i)^j$ has a 3-power denominator at most 3^{bij} , and

$$(u^i)^x = \left(\frac{M^a}{N^b} \right)^i$$

has an N -power denominator N^{bi} . Clearing rows gives the explicit upper budget

$$\mathfrak{d}_S \leq 3^{bd \binom{d+2}{2}} N^{b \binom{d+2}{2}} \times (\text{basis-unit factors}). \quad (50)$$

For fixed d , this is exponential in b , while (49) is a negative power of b . Therefore the direct “choose many group elements close to 1” strategy cannot work with termwise denominator clearing.

Proposition 10.2 (One-ratio cluster no-go). *Any proof based only on:*

1. a single ratio $u = 2^a 3^{-b} \rightarrow 1$;
2. the geometric nodes $1, u, \dots, u^{d+1}$ with fixed d ;
3. rowwise or termwise clearing at primes dividing $6MN$;

has an exponential denominator budget and only polynomial Archimedean decay. Those termwise estimates cannot certify (48) for large b .

The proposition does not rule out cancellation in the determinant. On the contrary, it identifies cancellation as the only surviving mechanism in this localized framework.

10.4 The finite-place cancellation target

For a chosen cluster, expand

$$\Delta_{d,x}^\times(E) = \sum_{\sigma} \operatorname{sgn}(\sigma) T_{\sigma}$$

into determinant terms. The automatic valuation lower bound is the minimum of the $v_p(T_{\sigma})$. A proof needs a strict gain:

$$v_p\left(\sum_{\sigma} \operatorname{sgn}(\sigma) T_{\sigma}\right) > \min_{\sigma} v_p(T_{\sigma}) \quad (51)$$

at enough primes in S , or a structural reason that the apparent denominator cancels before the determinant is evaluated.

This is a finite-prime, output-sensitive target. The input factorial has already extracted all universal unramified divisibility, and Proposition 10.2 has removed termwise structural clearing. What remains is not another counting estimate; it is the arithmetic of the initial forms of one specific determinant.

Status. The localized factorial, finite-support criterion, and one-ratio no-go are [\[proved here\]](#). The cancellation inequality (51) is [\[open target\]](#) and is one of the highest-priority bridges identified in this report.

11 A general local calculus for two-generator semigroups

The input computation extends to any pair of positive integers. This is needed because the output generators M, N are unknown.

Let

$$S(A, B) = \{A^a B^b : a, b \in \mathbb{N}_0\}.$$

11.1 Unramified odd primes

Theorem 11.1 (General unramified orbit formula). *Let p be odd and $p \nmid AB$. Put*

$$h_{A,B;p}(e) = |\langle A, B \rangle \subset (\mathbb{Z}/p^e\mathbb{Z})^\times|.$$

Then

$$v_p(k!_{S(A,B)}) = \sum_{e \geq 1} \left\lfloor \frac{k}{h_{A,B;p}(e)} \right\rfloor. \quad (52)$$

Proof. The proof of Theorem 3.1 uses only that the two generators are p -adic units and that $\mathbb{Z}_p^\times$ is procyclic. $\square$

11.2 Ramified primes as rooted merge systems

When $p \mid AB$, partition the closure by exact p -adic valuation. Distinct valuation branches have a predictable cross-difference valuation:

$$v_p(x - y) = \min(v_p(x), v_p(y)) \quad \text{when } v_p(x) \neq v_p(y).$$

Within a fixed branch, divide by the common power of p ; what remains is a compact unit orbit or a rescaled copy of the whole closure. The global p -ordering is therefore a greedy merge of local cost streams.

For $(A, B) = (2, 3)$ at $p = 2$, there are exactly two self-similar branches and the merge is Theorem 3.3. For a general output pair, there may be several branches or both generators may be ramified. When exactly one generator is ramified, Section 14 proves a closed two-stream recursion. When both are ramified, overlap between the scaled branches is the unresolved finite-state problem isolated below.

Research problem 11.2 (Output-prime tomography). *For a hypothetical pair (M, N) , compute the ramified merge systems at every $p \mid MN$, and relate their renewal slopes to:*

$$v_p(M), \quad v_p(N), \quad \overline{\langle Mp^{-v_p(M)}, Np^{-v_p(N)} \rangle} \subset \mathbb{Z}_p^\times.$$

Determine whether the collection of these slopes is compatible with an order-preserving power map from $\mathcal{B}$.

This is a finite calculation once M, N are specified. A proof would need a uniform inequality valid for every possible output pair, or a descent reducing the possible valuation profiles.

11.3 A useful separation of roles

The two-generator local calculus separates three ingredients that are often conflated.

Orbit size. At unramified primes, this is $h_{A,B;p}(e)$.

Valuation branching. At ramified primes, this records how many scaled copies or unit orbits occur at each depth.

Global synchronization. A regular basis exists primewise, but a single sequence of nodes may fail to be optimal at all primes; Theorem 8.1 detects the first failure for $(2, 3)$.

Any proposed determinant proof can now be audited against these three layers.

12 The first regular-basis element that is not a Newton product

The failure of a simultaneous ordering at six nodes does not destroy the regular basis. It forces the local optimal polynomials to be glued coefficientwise. This section makes that repair completely explicit.

For degrees at most five one may use the nested node sets

$$\emptyset, \quad \{1\}, \quad \{1, 2\}, \quad \{1, 2, 3\}, \quad \{1, 2, 3, 4\}, \quad \{1, 2, 3, 4, 6\}.$$

Their successive factorial defects are one. Hence the polynomials

$$F_k^N(T) = \frac{\prod_{a \in E_k} (T - a)}{k!_{\mathcal{B}}} \quad (0 \leq k \leq 5) \quad (53)$$

are integer-valued on $\mathcal{B}$ and have the optimal leading coefficients. At degree six no six-element node set supplies such a global Newton numerator, by Theorem 8.1.

Theorem 12.1 (An explicit degree-six regular-basis element). *Put*

$$P_6(T) = T^6 + 840T^5 + 175T^4 - 1260T^3 \quad (54)$$

$$- 2156T^2 + 1680T + 720. \quad (55)$$

Then

$$F_6(T) = \frac{P_6(T)}{5040} \in \text{Int}(\mathcal{B}, \mathbb{Z}), \quad \text{lc}(F_6) = \frac{1}{6!_{\mathcal{B}}}. \quad (56)$$

Consequently

$$1, F_1^N, \dots, F_5^N, F_6$$

is a regular basis through degree six.

Proof. The denominator is

$$6!_{\mathcal{B}} = 2^4 3^2 5 7 = 5040.$$

The polynomial in (55) is the coefficientwise Chinese-remainder gluing of four monic local numerators:

$$P_6(T) \equiv (T-1)(T-2)(T-3)(T-4)(T-6)(T-8) \pmod{16}, \quad (57)$$

$$P_6(T) \equiv T(T-1)(T-2)(T-3)(T-4)(T-5) \pmod{9}, \quad (58)$$

$$P_6(T) \equiv T^2(T^4 - 1) \pmod{5}, \quad (59)$$

$$P_6(T) \equiv T^6 - 1 \pmod{7}. \quad (60)$$

For $q \in \mathcal{B}$, the possible residues modulo 16 are

$$0, 1, 2, 3, 4, 6, 8, 9, 11, 12;$$

direct substitution in the product in (57) gives zero modulo 16. The product of six consecutive integers in (58) is divisible by $6!$, hence by 9. At the prime 5, every $q \in \mathcal{B}$ is a unit and satisfies $q^4 \equiv 1 \pmod{5}$. At the prime 7, it is a unit and satisfies $q^6 \equiv 1 \pmod{7}$. Thus every prime-power factor of 5040 divides $P_6(q)$, so $F_6(q) \in \mathbb{Z}$. Its leading coefficient is $1/5040 = 1/6!_{\mathcal{B}}$, which is optimal by the characteristic-ideal theorem. Adjoining it to any regular basis through degree five gives the asserted basis. $\square$

Remark 12.2 (What the factor seven actually means). The obstruction in Theorem 8.1 is not a missing denominator. The optimal denominator 5040 exists and the polynomial (55) realizes it. What fails is the stronger demand that one monic numerator factor into six linear terms with all six roots in $\mathcal{B}$. The first global incompatibility is therefore a coefficient-gluing defect, not a failure of integer-valued interpolation.

Status. Theorem 12.1 is [\[proved here\]](#). Its four congruences and the displayed CRT coefficients are also checked by the exact script in Appendix A.

13 Optimal factorial energies through eight nodes

The first-defect theorem can be pushed two steps farther. This produces a sharp finite picture of how quickly global synchronization deteriorates.

For $m \geq 1$, define

$$\mu_m(\mathcal{B}) = \min_{\substack{E \subset \mathcal{B} \\ |E|=m}} Q_{\mathcal{B}}(E). \quad (61)$$

The minimum exists because the Vandermonde product tends to infinity with the diameter of an integer set.

13.1 A general span bound

Lemma 13.1 (Integer-span energy bound). *Let $0 = y_0 < y_1 < \cdots < y_{m-1} = L$ be distinct integers, with $m \geq 3$. Then*

$$V(y_0, \dots, y_{m-1}) \geq \left(\prod_{j=1}^{m-3} j! \right) L(L-1)^{m-2}. \quad (62)$$

Proof. The Vandermonde product of the $m-2$ interior integers is minimized by consecutive integers and is therefore at least $\prod_{j=1}^{m-3} j!$. The endpoint pair contributes L , and for every interior point y_i one has $y_i(L - y_i) \geq L - 1$. Multiplication gives the claim. $\square$

13.2 The exact minima

Theorem 13.2 (First eight optimal energies). *The exact defect minima are*

$$\boxed{(\mu_1, \mu_2, \dots, \mu_8) = (1, 1, 1, 1, 1, 7, 7, 396)}. \quad (63)$$

For $m = 6, 7, 8$, the minimizers are unique and equal respectively to

$$E_6 = \{1, 2, 3, 4, 6, 8\}, \quad (64)$$

$$E_7 = \{1, 2, 3, 4, 6, 8, 9\}, \quad (65)$$

$$E_8 = \{1, 2, 3, 4, 6, 8, 9, 12\}. \quad (66)$$

Their defects are

$$Q_{\mathcal{B}}(E_6) = 7, \quad Q_{\mathcal{B}}(E_7) = 7, \quad Q_{\mathcal{B}}(E_8) = 396.$$

Proof. The values through $m = 5$ follow from the Newton prefixes in (53). Theorem 8.1 proves the value at six. To see uniqueness, the equality equation

$$V(E) = 7 \prod_{j=0}^5 j!_{\mathcal{B}}$$

and Lemma 13.1 restrict the normalized gap pattern to

$$(0, 1, 2, 3, 5, 7) \quad \text{or its reflection} \quad (0, 2, 4, 5, 6, 7).$$

The first contains four consecutive integers. The only four consecutive elements of $\mathcal{B}$ are 1, 2, 3, 4: this follows immediately from the classification of consecutive triples in the proof of Theorem 8.1. Hence its translate is exactly E_6 . The reflected pattern would place its four-consecutive run at a positive offset and has no positive translate in $\mathcal{B}$.

For seven nodes,

$$D_7 := \prod_{j=0}^6 j!_{\mathcal{B}} = 1741824000.$$

If $Q_{\mathcal{B}}(E) < 7$, Lemma 13.1 gives $L \leq 19$. Exhausting the $\binom{19}{6} = 27132$ normalized patterns leaves, up to reflection, only

Q	pattern
2	(0, 1, 3, 4, 5, 6, 8)
3	(0, 1, 2, 3, 5, 6, 8)
3	(0, 1, 2, 3, 4, 5, 10).

Every pattern contains four consecutive integers; forcing that run to be 1, 2, 3, 4 makes one of the remaining entries a non-smooth integer. Thus $Q \geq 7$. Equality leaves four normalized patterns, paired by reflection:

$$\begin{aligned} (0, 1, 2, 3, 5, 7, 8), & \quad (0, 1, 2, 3, 6, 7, 8), \\ (0, 1, 2, 5, 6, 7, 8), & \quad (0, 1, 3, 5, 6, 7, 8). \end{aligned}$$

Only the first has a positive smooth translate, namely E_7 .

For eight nodes,

$$D_8 := \prod_{j=0}^7 j!_{\mathcal{B}} = 17557585920000.$$

The set E_8 has

$$V(E_8) = 396D_8.$$

If a smaller defect existed, Lemma 13.1 would force $L \leq 42$. There are exactly

$$\sum_{L=7}^{42} \binom{L-1}{6} = \binom{42}{7} = 26978328$$

normalized integer patterns. Exact integer filtering by $0 < V(E)/D_8 < 396$ leaves only 92, with the defect histogram printed in Table 3. Of these, 74 contain four consecutive offsets. A positive smooth translate is then possible only when the run starts at offset zero and the translation is one; direct evaluation of the remaining offsets eliminates all 17 such patterns. The other 18 patterns contain a consecutive triple. Its translated starting value must be one or two. Thus, for a triple beginning at normalized offset r , the only candidate translations are $t = 1 - r$ and $t = 2 - r$, retained only when $t > 0$; in particular only $r = 0$ or $r = 1$ can contribute. Every such candidate is checked exactly and fails. Therefore no defect below 396 occurs.

At equality, exactly eight normalized patterns survive the Vandermonde equation. The same consecutive-run test eliminates seven; the sole smooth translate is

$$1 + (0, 1, 2, 3, 5, 7, 8, 11) = E_8.$$

The complete enumeration and all membership checks are reproduced by the verifier in Appendix B. $\square$

Corollary 13.3 (The synchronization collapse is abrupt). *A globally optimal node set exists through five nodes, incurs only one factor 7 at six and seven nodes, and then pays at least*

$$396 = 2^2 \cdot 3^2 \cdot 11$$

at eight nodes. The defect is therefore not governed by a gradual loss at one fixed prime. It is a genuinely global compatibility cost.

Defect	Count	Defect	Count	Defect	Count
2	2	3	2	12	2
15	2	18	6	20	2
22	2	30	2	33	4
60	4	63	2	77	2
78	2	84	2	99	2
105	2	108	4	110	4
128	2	132	4	162	2
180	2	198	2	220	2
240	2	252	6	270	4
286	4	288	2	297	2
308	2	324	2	330	4
378	2				

Table 3: Exact histogram of normalized eight-point integer patterns with defect below 396. None has a translate contained in $\mathcal{B}$.

Status. Lemma 13.1 is [\[proved here\]](#). Theorem 13.2 combines that proof with [\[exact computation\]](#) finite certificates. The search is exhaustive because the span bounds are proved before enumeration; no empirical height cutoff is used.

14 A closed ramified recursion for one structural generator

The two-adic recursion is not peculiar to $(2, 3)$. It is the universal local model when exactly one generator is divisible by the prime.

Theorem 14.1 (One-ramified-generator merge). *Let $A, B > 1$, let $p \mid A$ and $p \nmid B$, and write*

$$A = p^r u, \quad r = v_p(A) \geq 1, \quad p \nmid u.$$

Let

$$S = S(A, B), \quad H = \overline{\langle B \rangle}^p \subset \mathbb{Z}_p^\times,$$

and put

$$a_k = \nu_{S,p}(k), \quad b_k = \nu_{H,p}(k).$$

Then, as ordered multisets,

$$\boxed{\{a_k : k \geq 0\} = \{rj + a_j : j \geq 0\} \uplus \{b_j : j \geq 0\}.} \quad (67)$$

Equivalently, the p -ordering is obtained by greedily merging a scaled-copy stream $rj + a_j$ and the unit-orbit stream b_j .

Proof. Let $K = \overline{S}^p$. Splitting according to whether the exponent of A is zero gives the disjoint decomposition

$$K = H \sqcup p^r u K. \quad (68)$$

Every cross-branch difference is a unit, hence contributes zero valuation. With j previous points in the second branch, multiplication by $p^r u$ contributes rj and reduces the internal problem to K , giving cost $rj + a_j$. With j previous unit points, the cost is b_j . The global greedy rule is their ordered merge. $\square$

Corollary 14.2 (Closed renewal slope). *Suppose $b_k/k \rightarrow \beta > 0$. Then*

$$\lim_{k \rightarrow \infty} \frac{a_k}{k} = c(r, \beta) := \frac{\sqrt{r^2 + 4r\beta} - r}{2}. \quad (69)$$

Proof. The counting-function proof of Lemma 3.4 applies with the first stream having slope $r + c$. Thus

$$\frac{1}{c} = \frac{1}{r + c} + \frac{1}{\beta}.$$

This is equivalent to $c(r + c) = r\beta$, whose positive root is (69). $\square$

For the input pair at $p = 2$, one has $r = 1$ and $\beta = 5/2$, recovering Corollary 3.5. For a hypothetical output pair (M, N) , every prime with $p \mid M$ and $p \nmid N$ has the explicit factorial slope

$$\lim_{k \rightarrow \infty} \frac{v_p(k!_C)}{k} = c \left(v_p(M), \sum_{e \geq 1} \frac{1}{\text{ord}_{p^e}(N)} \right) \quad (70)$$

for odd p , with the evident closed-orbit interpretation at $p = 2$.

Remark 14.3. The unit factor $u = A/p^r$ disappears from the renewal law. It rotates the scaled copy by a unit but does not change any difference valuation. Thus an asymmetric support prime is completely controlled by one ramification index and one unit-orbit density.

Research problem 14.4 (Shared-support renewal). *When $p \mid A$ and $p \mid B$, the two scaled copies overlap and (68) is no longer disjoint. Construct a finite min-plus renewal system from the numerical semigroup generated by $v_p(A)$ and $v_p(B)$, including the unit cosets, and extract its asymptotic slope and first correction term.*

Status. The merge theorem and the closed slope are [\[proved here\]](#). They turn the previously qualitative “output-prime tomography” proposal into an exact formula at every prime dividing exactly one output generator. The shared-support case remains [\[open target\]](#).

15 What the new factorial calculus changes in the proof search

The preceding results do not prove Conjecture 1.1. They do, however, change the shape of several proof attempts in a precise way.

15.1 What is now exact

A hypothetical counterexample supplies an integral column (q_i^x) on every input node set. The optimal universal denominator of the polynomial columns is no longer an unknown Smith factor: it is

$$\prod_{j=0}^d j!_{\mathcal{B}},$$

with every local exponent explicit. The first non-Newton basis element is explicit, and the best global node energies are known through eight points. At any asymmetric output prime, the output factorial has the closed renewal slope (70).

These facts remove three ambiguities from determinant experiments:

1. a small determinant may no longer be credited to a denominator that is merely an artifact of the monomial basis;
2. a node set may be scored by its exact defect $Q_{\mathcal{B}}(E)$, rather than by raw pairwise gaps;
3. output-prime gains can be separated into automatic renewal mass and genuine signed cancellation.

15.2 What remains genuinely missing

After optimal normalization, the integer determinant still contains the factor

$$\binom{x}{d+1} \xi^{x-d-1}.$$

For the known counterexample-free range, a hypothetical x is already large. Fixed-degree curvature therefore grows rather than shrinks on positive integer nodes. Increasing the degree introduces more factorial mass, but Theorem 13.2 shows that globally compatible node sets pay an additional arithmetic energy that rapidly ceases to be one.

Localization removes the two input structural primes from the factorial, but Proposition 10.2 shows that a one-ratio cluster then pays an exponential output denominator. The only unexhausted mechanism is cancellation in the structural-prime initial forms of the normalized determinant. The factorial calculus identifies this as the frontier instead of another undifferentiated “try a larger determinant” instruction.

15.3 A sharpened conditional criterion

For a localized determinant $\Delta^\times$, write at each structural prime

$$\Delta^\times = p^{m_p} (I_p + pJ_p),$$

where m_p is the minimum term valuation and I_p is the signed initial form over $\mathbb{F}_p$. The automatic factorial and row-clearing contributions determine m_p . The missing gain is exactly the vanishing of I_p .

Proof criterion 15.1 (Initial-form bridge). *Conjecture 1.1 follows if one can choose a family of localized regular-basis node sets for which*

1. *the Archimedean determinant has logarithm $-A_d + o(A_d)$ with $A_d \rightarrow \infty$;*
2. *at primes $p \mid 6MN$, the initial forms vanish to total weighted depth exceeding the termwise denominator budget by $A_d/\log p + o(A_d)$;*
3. *the determinant is nonzero, for example by total positivity or a Chebyshev argument.*

This is a sufficient condition, not a claim that such a family exists. Its advantage is that every universal local contribution has already been subtracted. Only the output-specific initial forms remain.

16 A prioritized research program

16.1 Priority A: structural-prime initial forms

For small degrees, build the explicit regular basis by local CRT gluing as in Theorem 12.1. Evaluate the localized determinant symbolically in the valuation variables

$$v_p(M), \quad v_p(N), \quad Mp^{-v_p(M)}, \quad Np^{-v_p(N)}.$$

The first target is not a large valuation bound. It is a classification of when the initial form I_p vanishes. A useful theorem would say that simultaneous nonvanishing at all structural primes is incompatible with the real ordering slope $\log M / \log 2 = \log N / \log 3$.

16.2 Priority B: paired regular-basis determinants

Construct an input determinant for t^x and an output determinant for $t^{1/x}$, each in its own Bhargava regular basis. Their Archimedean divided differences have reciprocal curvatures, while their finite-place normalizations are governed by the two factorial spectra. The concrete target is an adelic product in which the leading structural denominators cancel before absolute values are taken. Separate inequalities for the two determinants are too weak.

16.3 Priority C: low-defect node classification

The exact sequence

$$1, 1, 1, 1, 1, 7, 7, 396$$

invites a systematic study of $\mu_m(\mathcal{B})$. The next useful outputs are:

1. a lower bound $\log \mu_m \gg m^2$ or a counterexample to it;
2. a structural description of minimizers in exponent-lattice coordinates;
3. a comparison between minimizers of $Q_{\mathcal{B}}$ and minimizers of the full determinant including the curvature factor.

The finite verifier already supports branch-and-bound extensions beyond eight nodes, but an unproved search cutoff must not be confused with a theorem.

16.4 Priority D: the global Stirling law

Resolve Problem 5.2. Even a conditional GRH theorem with a power-saving error would quantify how much denominator is available at moving exceptional-order primes. The result would be independently interesting as a rank-two generalized-factorial analogue of Artin-density problems.

16.5 Priority E: shared-support renewal

Complete Problem 14.4. If every prime dividing one output also divides the other, the shared-support systems are unavoidable. Their min-plus spectral radii should give a finite collection of exact slopes. Comparing those slopes with the Smith-normal form constraints of the output valuation matrix may force a support split or a descent.

16.6 Priority F: exact lattice optimization

For a finite node set E , the evaluation vectors

$$(F_j(q))_{q \in E}, \quad 0 \leq j \leq d,$$

form an explicit integral lattice. Compute its dual shortest vector and the distance of $(q^x)_{q \in E}$ symbolically in M, N . A dual vector with alternating signs is especially valuable because total positivity can certify nonvanishing. The correct computation is an exact Smith/Hermite-normal-form calculation, not floating-point least squares.

16.7 Priority G: formalization before scale

Formalize the local formulas, cutoff, CRT basis, and the six-to-eight node certificates before running larger searches. This prevents the research loop from accumulating another layer of untracked numerical evidence.

17 Lean-oriented formalization plan

No Lean file is claimed here. The following decomposition is intended to minimize the amount of general infrastructure needed for a kernel-checked version.

17.1 Module 1: local orbit counts

Represent the finite image of $\mathcal{B}$ modulo p^e as the subgroup generated by 2 and 3. Prove the level-count identity

$$v_p \left(\prod_i (y - a_i) \right) = \sum_{e \geq 1} \# \{i : y \equiv a_i \pmod{p^e}\}$$

for nonzero products. The odd-prime orbit theorem then reduces to finite cyclic-group counting.

Suggested theorem interfaces are:

```
theorem pseq_two_three_of_prime_ge_five
  (hp : 5 <= p) :
  pseq twoThree p k =
    Finset.sum (contributingLevels p k)
      (fun e => k / subgroupCard p e)

theorem subgroupCard_eq_lcm_order :
  subgroupCard p e =
    Nat.lcm (orderOfMod (2 : ZMod (p^e)))
      (orderOfMod (3 : ZMod (p^e)))
```

17.2 Module 2: structural primes

At 3, formalize surjectivity of $2^a 3^b$ modulo 3^e . At 2, define the two cost streams and the deterministic tie-broken merge. Prove that the merge output satisfies the greedy minimum directly; Bhargava’s full choice-independence theorem is not needed for this specific set if the local minimum is verified.

17.3 Module 3: finite support and exact factorization

Formalize Theorem 4.1 using the $(R + 1)^2$ exponent box and the pigeonhole principle. A certified factorization for fixed k can then be expressed as a finite product over primes below the proved cutoff. The tables should be generated from evaluated theorems, not imported as opaque constants.

17.4 Module 4: integer-valued CRT basis

For P_6 , prove the four coefficient congruences (57)–(60) and the finite residue classification modulo 16. The conclusion

$$5040 \mid P_6(2^a 3^b)$$

is elementary modular arithmetic and is an excellent first formal regular-basis certificate.

17.5 Module 5: defect certificates

Encode normalized gap patterns as strictly increasing finite vectors. The span lemma reduces each theorem to a finite list. For six and seven nodes, `native_decide` or reflective normalization should suffice. For eight nodes, import a compact certificate containing the 92 surviving patterns and verify:

1. every omitted pattern has Vandermonde at least $396D_8$ or nonintegral normalized ratio;
2. every listed pattern fails the smooth-translate predicate;
3. the eight equality patterns have exactly one successful translate.

The generator of the certificate can remain external; the checker must be small and complete.

18 Reproducibility and trust boundary

18.1 Exact arithmetic only

All computations reported here use integers, modular exponentiation, exact multiplicative orders, and exact Vandermonde products. Floating-point numbers appear only when displaying logarithms or limiting constants. The factorial code uses the proved cutoff $p^e < 6^{\lfloor \sqrt{k} \rfloor}$; the defect code uses the proved span bounds.

18.2 Independent checks performed

The following checks were run in the build environment:

1. the merge recursion reproduces the first 51 two-adic exponents;
2. every displayed factorial is a multiple of the ordinary factorial;
3. direct greedy searches on finite exponent rectangles agree with the local orbit formula for small primes and degrees;
4. $P_6(q)$ is divisible by 5040 for a large rectangle of exponent pairs, in addition to the proof by congruences;

5. the six-, seven-, and eight-node Vandermonde values and defect denominators are recomputed from scratch;
6. the bounded smooth-node search independently finds E_6, E_7, E_8 as the first minimizers.

18.3 Public-breakthrough verification note

The competitive context in the prompt is partly publicly checkable. An official Anthropic research note states that a similar multi-agent prompt was used in work on the Jacobian counterexample, and subsequent arXiv papers give independent mathematical accounts of the counterexamples in dimensions at least three. A public rank-record page reports a curve of rank at least thirty found with Anthropic researchers. Public posts also report resolution of the previously unknown Hadamard orders below two thousand, although at the time of this report I did not locate one consolidated archival paper containing all twelve matrices and their verification certificates. Most importantly, I found no public Alaoglu–Erdős proof, preprint, or Lean repository that could be audited. None of these announcements is used as a mathematical premise in this report [12, 11, 13, 14].

Status. The source listings below are the exact programs used for the reported certificates, modulo line wrapping in the PDF. The claims about public announcements are contextual and [external theorem]; the mathematical results of this report do not depend on them.

19 Conclusions

The Alaoglu–Erdős conjecture remains unproved here. The new progress is a completed local and finite arithmetic layer that the supplied report had left open.

For the semigroup $\mathcal{B} = \langle 2, 3 \rangle_+$, every unramified local factorial is now an exact orbit sum, the three-adic factorial is ordinary, and the two-adic factorial is a solvable self-similar renewal with slope $(\sqrt{11} - 1)/2$. A square-box collision theorem makes the global factorization finite and certifiable. The first non-Newton regular-basis polynomial is explicit. The optimal global Vandermonde defects through eight nodes are exactly

$$1, 1, 1, 1, 1, 7, 7, 396,$$

with unique minimizers at the last three stages. At every prime dividing exactly one generator of a two-generator semigroup, the ramified factorial has a closed merge law and a closed asymptotic slope.

These results sharpen the remaining obstruction. Universal unramified divisibility is no longer missing; it is completely normalized away. Dense one-ratio clusters do not work under termwise structural clearing. What remains is signed, output-sensitive cancellation at the finitely many primes dividing $6MN$, preferably in a paired input/output determinant. That is a narrower and more falsifiable target than the generic determinant amplification programs considered before.

The immediate proof-producing next step is therefore concrete: construct the regular basis in several more degrees, compute the structural-prime initial forms symbolically, and classify their vanishing in terms of the output valuation matrix. Either a uniform nonvanishing theorem will close this route decisively, or a systematic cancellation pattern will supply the missing adelic gain.

A Exact factorial and CRT verifier

The following Python program reproduces the factorial tables, the local merge, the CRT polynomial, and the small bounded checks. It requires Python 3 and SymPy.

```

1 from __future__ import annotations
2 from collections import Counter
3 from math import factorial, isqrt, lcm, log
4 from sympy import Poly, expand, factorint, n_order, primerange, symbols
5 from sympy.ntheory.modular import crt
6
7
8 def vp_factorial(n: int, p: int) -> int:
9     total = 0
10    while n:
11        n //= p
12        total += n
13    return total
14
15
16 def unit_cost_two(k: int) -> int:
17     return k + k // 2 + vp_factorial(k, 2)
18
19
20 def two_adic_sequence(nmax: int) -> list[int]:
21     # Start with 1 in the odd branch U.
22     a = [0]
23     even_count, odd_count = 0, 1
24     for _stage in range(1, nmax + 1):
25         even_cost = even_count + a[even_count]
26         odd_cost = unit_cost_two(odd_count)
27         a.append(min(even_cost, odd_cost))
28         if even_cost <= odd_cost: # deterministic tie break
29             even_count += 1
30         else:
31             odd_count += 1
32     return a
33
34
35 def bhargava_factorization(k: int, a2: list[int]) -> dict[int, int]:
36     f: Counter[int] = Counter()
37     if a2[k]:
38         f[2] = a2[k]
39     e3 = vp_factorial(k, 3)
40     if e3:
41         f[3] = e3
42     if k < 2:
43         return dict(f)
44
45     cutoff = 6 ** isqrt(k)
46     for p in primerange(5, cutoff):
47         pe = p
48         while pe < cutoff:
49             h = lcm(int(n_order(2, pe)), int(n_order(3, pe)))
50             if h <= k:
51                 f[p] += k // h
52             pe *= p
53     return dict(sorted(f.items()))
54
55
56 def value(factors: dict[int, int]) -> int:
57     z = 1
58     for p, e in factors.items():
59         z *= p ** e
60     return z
61
62
63 def vandermonde(nodes: tuple[int, ...]) -> int:

```

```

64     z = 1
65     for i in range(len(nodes)):
66         for j in range(i + 1, len(nodes)):
67             z *= nodes[j] - nodes[i]
68     return z
69
70
71 def product_factorials(m: int, a2: list[int]) -> int:
72     z = 1
73     for k in range(m):
74         z *= value(bhargava_factorization(k, a2))
75     return z
76
77
78 def crt_polynomial_degree_six():
79     T = symbols("T")
80     moduli = [16, 9, 5, 7]
81     local = [
82         expand((T-1)*(T-2)*(T-3)*(T-4)*(T-6)*(T-8)),
83         expand(T*(T-1)*(T-2)*(T-3)*(T-4)*(T-5)),
84         expand(T**2*(T**4-1)),
85         expand(T**6-1),
86     ]
87     coeff = []
88     for j in range(7):
89         residues = [int(Poly(P, T).nth(j)) % m
90                     for P, m in zip(local, moduli)]
91         x, modulus = crt(moduli, residues)
92         x, modulus = int(x), int(modulus)
93         if x > modulus // 2:
94             x -= modulus
95         coeff.append(x)
96     return expand(sum(coeff[j] * T**j for j in range(7)))
97
98
99 def main() -> None:
100     a2 = two_adic_sequence(50)
101     assert a2[:21] == [
102         0, 0, 1, 1, 3, 4, 4, 5, 7, 9, 9,
103         10, 10, 12, 13, 14, 15, 18, 19, 19, 20,
104     ]
105     for k in range(51):
106         f = bhargava_factorization(k, a2)
107         assert value(f) % factorial(k) == 0
108
109     P6 = crt_polynomial_degree_six()
110     assert str(P6) == (
111         "T**6 + 840*T**5 + 175*T**4 - 1260*T**3 "
112         "- 2156*T**2 + 1680*T + 720"
113     )
114
115     expected = {
116         6: ((1, 2, 3, 4, 6, 8), 7),
117         7: ((1, 2, 3, 4, 6, 8, 9), 7),
118         8: ((1, 2, 3, 4, 6, 8, 9, 12), 396),
119     }
120     for m, (nodes, defect) in expected.items():
121         D = product_factorials(m, a2)
122         assert vandermonde(nodes) == defect * D
123
124     print("P6 =", P6)
125     for k in (20, 25, 30, 35, 40, 45, 50):
126         f = bhargava_factorization(k, a2)
127         z = value(f)
128         print(k, z // factorial(k), f, log(z // factorial(k)))
129
130
131 if __name__ == "__main__":
132     main()

```

B Exhaustive defect verifier through eight nodes

For speed, the eight-node normalized-pattern exhaustion was performed in C++ using signed 128-bit integer arithmetic. The maximum Vandermonde below the threshold is far below 2^{127} . The program emits every pattern with integral defect below 396 and every pattern at equality; the short smooth-translate checker then applies the consecutive-run classification described in the proof.

```

1 #include <algorithm>
2 #include <array>
3 #include <cstdint>
4 #include <iostream>
5 #include <map>
6 #include <vector>
7 using i128 = __int128_t;
8
9 static i128 vandermonde(const std::array<int, 8>& x) {
10     i128 v = 1;
11     for (int i = 0; i < 8; ++i)
12         for (int j = i + 1; j < 8; ++j)
13             v *= (x[j] - x[i]);
14     return v;
15 }
16
17 static bool smooth23(std::int64_t n) {
18     if (n <= 0) return false;
19     while (n % 2 == 0) n /= 2;
20     while (n % 3 == 0) n /= 3;
21     return n == 1;
22 }
23
24 static bool has_consecutive_triple(const std::array<int, 8>& x) {
25     for (int i = 0; i + 2 < 8; ++i)
26         if (x[i+1] == x[i] + 1 && x[i+2] == x[i] + 2) return true;
27     return false;
28 }
29
30 static bool has_smooth_translate(const std::array<int, 8>& x) {
31     // A consecutive triple can only translate to {1,2,3} or {2,3,4}.
32     for (int i = 0; i + 2 < 8; ++i) {
33         if (x[i+1] == x[i] + 1 && x[i+2] == x[i] + 2) {
34             for (int start : {1, 2}) {
35                 const int t = start - x[i];
36                 if (t <= 0) continue;
37                 bool ok = true;
38                 for (int d : x) ok = ok && smooth23(t + d);
39                 if (ok) return true;
40             }
41         }
42     }
43     return false;
44 }
45
46 int main() {
47     constexpr std::int64_t D8 = 17557585920000LL;
48     std::map<long long, int> histogram;
49     int below = 0, equal = 0, smooth_equal = 0;
50
51     for (int L = 7; L <= 42; ++L)
52         for (int a = 1; a < L; ++a)
53             for (int b = a + 1; b < L; ++b)
54                 for (int c = b + 1; c < L; ++c)
55                     for (int d = c + 1; d < L; ++d)
56                         for (int e = d + 1; e < L; ++e)
57                             for (int f = e + 1; f < L; ++f) {
58                                 std::array<int, 8> x{0, a, b, c, d, e, f, L};
59                                 const i128 V = vandermonde(x);
60                                 if (V > i128(396) * D8 || V % D8 != 0) continue;

```

```

61     if (!has_consecutive_triple(x)) {
62         std::cerr << "survivor without consecutive triple\n";
63         return 3;
64     }
65     const long long q = static_cast<long long>(V / D8);
66     if (q < 396) {
67         ++below;
68         ++histogram[q];
69         if (has_smooth_translate(x)) {
70             std::cerr << "unexpected lower translate\n";
71             return 1;
72         }
73     } else {
74         ++equal;
75         if (has_smooth_translate(x)) ++smooth_equal;
76     }
77 }
78
79 if (below != 92 || equal != 8 || smooth_equal != 1) return 2;
80 for (auto [q, count] : histogram)
81     std::cout << q << " " << count << "\n";
82 }

```

The six- and seven-node searches are identical with fewer loops and the denominators 345600 and 1741824000. Their complete pattern counts are 792 and 27132, respectively.

C Selected local orbit data

For the first unramified primes, no delayed lifting occurs: $c_p = 1$ in (7). Thus

$$\alpha_p = \frac{p}{h_p(1)(p-1)}.$$

p	$h_p(1)$	α_p	δ_p
5	4	5/16	1/16
7	6	7/36	1/36
11	10	11/100	1/100
13	12	13/144	1/144
17	16	17/256	1/256
19	18	19/324	1/324
23	11	23/242	6/121
29	28	29/784	1/784
31	30	31/900	1/900
47	23	47/1058	12/529

Table 4: Orbit sizes and fixed-prime factorial excess. The primes 23 and 47 display proper two-generator orbit subgroups modulo p .

D Dependency and status matrix

Statement	Status	Dependency
Odd-prime orbit factorial formula	[proved here]	Closure invariance; cyclic units
Three-adic ordinary factorial	[proved here]	Primitive-root property of 2
Two-adic merge recursion	[proved here]	Self-similar closure split
Two-adic slope $(\sqrt{11} - 1)/2$	[proved here]	Renewal counting lemma
Square-box collision cutoff	[proved here]	Pigeonhole principle
Exact factorial tables through 50	[exact computation]	Cutoff plus modular orders
Rank-two Stirling law	[open target]	Average small-order primes
Normalized determinant integrality	[proved here]	Regular basis; Chebyshev system
First defect 7 at six nodes	[proved here]	Span bound; finite certificate
Exact minima through eight nodes	[exact computation]	Span bounds; exhaustive certificates
Explicit degree-six regular basis	[proved here]	CRT congruences
One-ramified-generator slope	[proved here]	Two-stream renewal
Shared-support renewal	[open target]	Overlapping valuation branches
Localized finite-support criterion	[conditional criterion]	Structural-prime denominator control
Initial-form cancellation bridge	[conditional criterion]	Output-sensitive cancellation
Full Alaoglu–Erdős conjecture	[open target]	No bridge yet established

E Literature and source notes

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